LAM NGO

Email • Google Scholar • LinkedIn • Website • Github

Research Interests: 3D Computer Vision, Spatial Intelligence, AI for Science & Health

EDUCATION

Master of Science in Computer Science **09/2023 - 12/2024**
City St George's, University of London • ***GPA: Distinction*** *London, UK*

- Thesis: Feature Extraction from Deep Dynamic Network for Reinforcement Learning.
- **Global STEM Leadership Master's Scholarship**: 75% tuition award, given to 10 top applicants.

Bachelor of Science in Mathematics. Minor in Biology **08/2016 - 05/2020**
Sewanee: The University of the South • ***GPA: 3.46/4.0*** *TN, USA*

- **International Honors Scholarship**: partial tuition award.

PUBLICATIONS AND PRESENTATIONS

[1] Child, C. & **Ngo, L.** (2026). *DeeP-Mod: Deep Dynamic Programming Based Environment Modelling Using Feature Extraction*. In: Jin, L., Wang, L. (eds) Advances in Neural Networks – ISNN 2025. ISNN 2025. Lecture Notes in Computer Science, vol 15951. Springer, Singapore. https://doi.org/10.1007/978-981-95-1233-1_3. *Preprint: arXiv:2504.20535*

- **Presenting author** at *19th International Symposium on Neural Networks (ISNN 2025)*, August 24, 2025, China

[2] Balogh, A., **Ngo, L.**, Zigler, K.S. et al. *Population genomics in two cave-obligate invertebrates confirms extremely limited dispersal between caves*. Sci Rep 10, 17554 (2020). https://doi.org/10.1038/s41598-020-74508-9

RESEARCH EXPERIENCE

Research Intern **12/2025 - Present**
Supervisor: Dr. Jiancheng Yang • Aalto University *Helsinki, Finland*
Simulation-Ready Cardiac Mesh – In Progress

- Develop methods for 3D reconstruction of cardiac structures via geometric distribution learning and diffusion posterior sampling, enabling anatomically accurate surface reconstruction from unseen sparse point clouds.

Research Engineer **03/2025 - 03/2026**
Supervisor: Dr. Giacomo Tarroni • CitAI Research Centre *London, UK*
Federated Dataset Simulation – In Progress

- Developed dataset partitioning pipelines using modern vision-language models (VLMs) and large language models (LLMs) to address the scarcity of real-world federated datasets for computer vision tasks.

Research Engineer **09/2024 - 08/2025**
Supervisor: Dr. Chris Child • City University London *London, UK*
DeeP-Mod – ISNN 2025

- Developed environment modeling techniques for reinforcement learning tasks by combining dynamic programming with feature extraction and Deep Q Networks, improving agent training in noisy settings.

Undergraduate Research Intern **02/2019 - 05/2020**
American Museum of Natural History *New York, USA*

- Worked as technical assistant for Terrapin Tracker, an IoT-based animal tracking system, which was selected as finalist for the Con X Tech Prize 2020.

INDUSTRY EXPERIENCE

C++ Programmer Mentee **10/2023 - 03/2024**
Ubisoft Leamington *Leamington Spa, UK*

- Built an entity-component game engine using modern C++ and internal graphics API.
- Implemented AI algorithms to anticipate player movements, enhancing gameplay difficulty.

Junior Programmer **06/2022 - 06/2023**
Indi Games Inc. *Ho Chi Minh City, Vietnam*

- Developed multiplayer game systems in Unity with real-time WebSocket integration, delivering three commercial products for Japanese clients.

Web Developer **03/2021 - 11/2021**
Teamscāl Pty Ltd *Ho Chi Minh City, Vietnam*

- Developed and deployed WordPress-based websites and plugins for Australian clients, leveraging HTML, CSS, and PHP for front-end and back-end integration.

CERTIFICATES

Deep Generative Learning & **Advanced Computer Vision** | DeepLearning.AI | 2025
Mathematics for Machine Learning: Linear Algebra | Coursera | 2018

SKILLS

Programming languages: Python, C++, TypeScript, C#
ML: PyTorch, TensorFlow, Keras, CUDA, OpenCV, NumPy, Pandas, Scikit-learn
Infrastructure: Docker, SLURM, Git, Linux, Multi-GPU/Multi-Node Training
Models: InternVL3 38B, LLaMA 3.3 70B, MiniCPM 2.6, GPT-o4-mini, CLIP, BLIP-2, Claude Code

REFERENCES

Dr. Jiancheng Yang • Assistant Professor & PI • Aalto University & ELLIS Institute Finland
Helsinki, Finland • Contact: jiancheng.yang@aalto.fi

Dr. Giacomo Tarroni • Associate Professor • City St George’s University of London
London, UK • Contact: giacomo.tarroni@citystgeorges.ac.uk